

Practice 5: Greedy Algorithm - Interval Scheduling and Interval Partitioning

Deadline: Apr.8/9/10. Must be checked onsite (during labs).Online checks via email or messaging apps are not accepted.

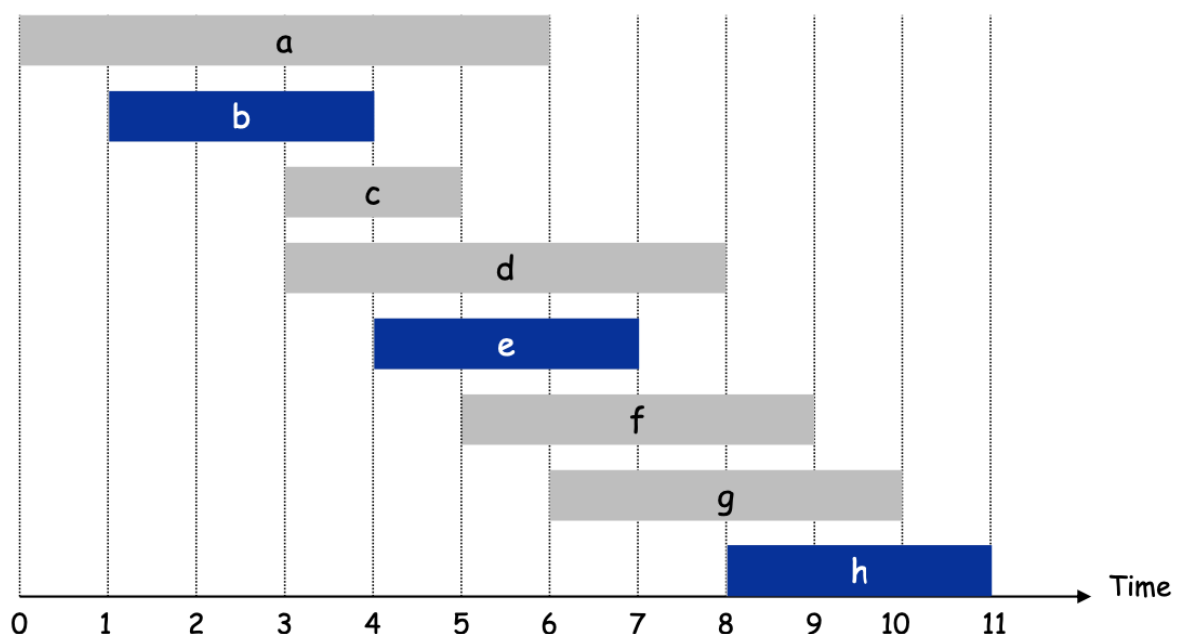
1. Objective

This Practice consists of two questions based on **Chapter 4.1.1 Interval Scheduling** and **Chapter 4.1.2 Interval Partitioning** in the lecture. It is intended to help students understand the basic concepts of greedy algorithms.

2. Tasks

2.1 Interval Scheduling

Given n jobs, each with a start time and an end time, find the maximum number of non-conflicting jobs that can be executed on a single thread.



Input Introduce

The first line contains an integer n , indicating that there are n jobs.

Each of the next n lines describes one of the jobs, containing two numbers separated by a space. These two numbers represent the job's start time and end time.

Output Introduce

Output a single line representing the maximum cardinality subset of mutually compatible jobs.

Input case1:

```
7
3 5
1 2
2 3
8 10
6 9
6 8
2 4
```

Output case1:

```
5
```

Input case2:

```
15
1 5
2 3
11 14
14 16
3 5
8 9
10 15
1 4
19 21
18 22
6 7
22 25
6 9
17 20
12 17
```

Output case2:

```
8
```

2.2 Interval Partitioning

In different time slots, there are several courses, each with a start time and an end time. Find the minimum number of classrooms needed to accommodate all courses without time conflicts.

Input Introduce:

The first line contains an integer **n**, indicating that there are n courses.

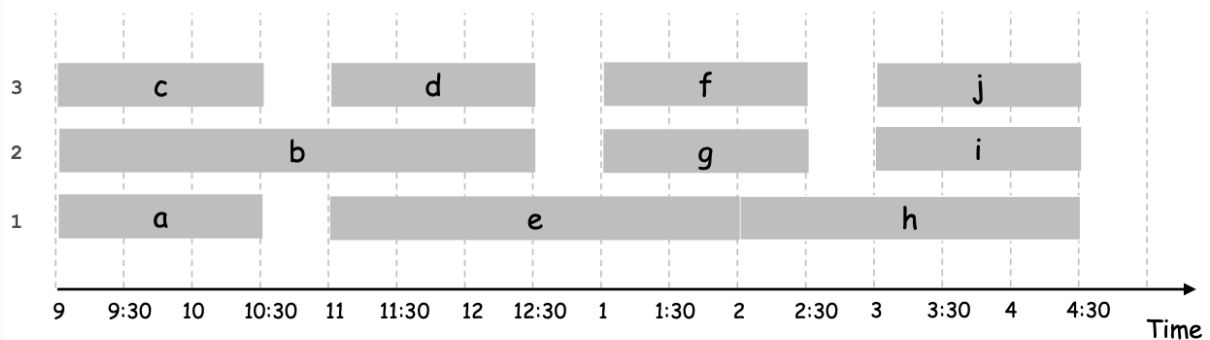
The next n lines each contain two floating-point numbers, separated by a space, representing the start time and end time of each course.

Output Introduce:

Output a single line representing the number of classrooms.

Input case1:

```
10
9 12.5
15 16.5
15 16.5
11 12.5
9 10.5
11 14
13 14.5
9 10.5
13 14.5
14 16.5
```



Output case1:

```
3
```

Input case2:

```
13
9 10.5
11 14
13 14.5
```

```
9 12.5
11 12.5
15 16.5
12 14
14 15.5
10 12.5
9 10.5
13 14.5
9 10.5
14 16.5
15 16.5
```

Output case2:

```
5
```

3. Submission

Show the following to your tutor or SA **onsite**:

Source Code: Well-commented code for both tasks.

Grading:

- Submissions on time: 0.25 points for each test cases per question.
- Late submissions: 0 points.